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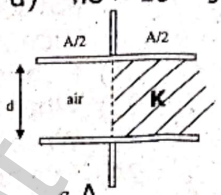
Virudhunagar District
Common Quarterly Examination - 2025

Standard 12
PHYSICS
PART - I

Time: 3.00 Hours Marks: 70

Answer all the questions **15x1=15**

1) Which charge configuration produces a uniform electric field?
   a) point charge b) uniformly charged infinite
   c) uniformly charged infinite plane d) uniformly charged spherical shell
2) Two points 'A' and 'B' are maintained at a potential of 7V and −4V respectively. The workdone in moving 50 electrons from A to B is
   a) 8.80×10^{-17} J b) -8.80×10^{-17} J c) 4.40×10^{-17} J d) -4.8×10^{-17} J
3) In a capacitor, the area of each plate is "A". Plates separation is "d". The space between the capacitor plates is filled with a dielectric of dielectric constant "k" as shown in figure.



   The value of capacitance of the capacitor is
   a) $\frac{\epsilon_0 A}{2d}(1+k)$ b) $\frac{\varepsilon_0 A}{d}(1+k)$ c) $\frac{\varepsilon_0 A}{2d}$ d) $\frac{\varepsilon_0 A}{2d}(1+2k)$
4) Two wires of 'A' and 'B' with circular cross - section are made up of the same material with equal length. Suppose $R_A = 3R_B$, then what is the ratio of radius of wire 'A' to that of 'B'?
   a) 3 b) $\sqrt{3}$ c) $\frac{1}{\sqrt{3}}$ d) $\frac{1}{3}$
5) A piece of copper and another of germanium are cooled from room temperature to 80k. The resistance of
   a) each of them increases b) copper increases and germanium decreases
   c) each of them decreases d) copper decreases and germanium increases
6) A circular coil of radius 5cm and 50 turns carries a current of 3A. The magnetic dipole moment of the coil is nearly
   a) 1 Am^2 b) 1.2 Am^2 c) 0.5 Am^2 d) 0.8 Am^2
7) The vertical component of Earth's magnetic field at a place is equal to the horizontal. What is the value of angle of dip at this place?
   a) 30° b) 45° c) 60° d) 90°
8) The unit of current reduction factor of the galvanometer is
   a) ampere b) ampere/radian c) radian/ampere d) no unit
9) When the current changes from +2A to −2A in 0.05s, and emf of 8V is induced in a coil. The co-efficient of self-induction of the coil is
   a) 0.2 H b) 0.4 H c) 0.8 H d) 0.1 H
10) In an oscillating LC circuit, the maximum charge on the capacitor is 'Q'. The charge on the capacitor when the energy is stored equally between the electric and magnetic fields is
   a) $Q/2$ b) $Q/\sqrt{3}$ c) $Q/\sqrt{2}$ d) Q
11) The mutual inductance between two coils
   a) doesnot depend on size and shape of the coils
   b) depends on relative orientation of the coils
   c) depends on current through either of the coils
   d) all the above
12) Which of the following is false for electro magnetic waves?
   a) transverse b) non-mechanical waves
   c) longitudinal d) produced by accelerating charges
13) Stars twinkle due to
   a) reflection b) total internal reflection
   c) refraction d) Polarisation
14) For light incident from air on a slab of refractive index 2, the maximum possible angle of refraction is
   a) 30° b) 45° c) 60° d) 90°

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15) The wavelength ratio of two light waves which undergo Rayleigh scattering is 5:3. The ratio of intensities of the scattered light is
a) 3 : 5 b) 5 : 3 c) 81 : 625 d) 625 : 81

PART - II

Answer any six questions. Q.No. 24 is compulsory. **6x2=12**

16) Define: Electrostatic potential at a point
17) A parallel plate capacitor has square plates of side 5 cm and separated by a distance of 1 mm. Calculate the capacitance of the capacitor
18) Mention any two applications of seebeck effect
19) State - "Ampere's circuital law"
20) Two materials x and y are magnetised whose values of intensity of magnetisation are 500 Am^{-1} and 2000 Am^{-1} respectively. If the magnetosing field is 1000Am^{-1}, then which one among these materials can be easily magnetised
21) Mention the ways of producing induced emf
22) What is displacement current?
23) What is the reason for reddish appearance of the sky during sunset and sunrise?
24) In a potentiometer arrangement, a cell of emf 1.25V gives a balance point at 35cm length of the wire. If the cell is replaced by another cell and the balance point shifts to 63cm. What is the emf of the second cell?

PART - III

Answer any six questions. Q.No. 33 is compulsory **6x3=18**

25) What are the difference between coulomb force and gravitational force?
26) State Kirchoff's current rule and voltage rule
27) How will you convert a Galvanometer into a voltmeter. Explain
28) Explain any one application of eddy currents
29) List out any six properties of electro magnetic waves
30) If the relative permeability are relative permittivity of a medium are 1.0 and 2.25 respectively, find the speed of the electro magnetic wave in this medium.
31) Derive the relation and focal length (f) and radius of curvature (R) for a spherical mirror
32) Find the dispersive power of a prism if the refractive indices of flint glass for red, green and violet colour are 1.613, 1.620 and 1.632 respectively
33) The self - inductance of an air - core solenoid is 4.8 mH. If its core is replaced by Iron core, then its self - inductance becomes 1.8H. Find out the relative permeability of iron

PART - IV

Answer all the questions. **5x5=25**

34) a) Obtain the expression for electric field due to an electric dipole at a point on its axial line

(OR)

b) Describe the Fizeau's method to determine the speed of light

35) a) i) How will you determine the internal resistance of a cell using voltmeter.
ii) A battery has an emf of 12V and connected to a resistor of 3Ω. The current in the circuit is 3.93 A. Calculate internal resistance of the cell

(OR)

b) Explain the principle construction and working of a Transformer.

36) a) Derive the expression for magnetic field at a ploint inside a current carrying 'toroid'

(OR)

b) i) Define Quality factor of a RLC series resonance circuit and obtain the expression for it.

ii) A 500 μH inductor, $\frac{80}{\pi^2}$ pF capacitor and 628 Ω resistor are connected to form a series RLC circuit. Calculate the quality factor at resonance.

37) a) Write and explain maxwell's equations in integral form

(OR)

b) Explain the principle construction and working of Van de Graff Generator

38) a) Obtain a relation for magnetic field at a point along the axis of a circular coil carrying current.

(OR)

b) Describe the Microscopic model of current and obtain general form of Ohm's law

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